

Sample protocol for spectral pigment measurements

Spectrophotometer, standard optics.

Three analysis periods with different spectrophotometers:

- Perkin Elmer Lambda 18, SN: 75009
- Perkin Elmer Lambda 650 S (Serial # 650L1806192)
- Shimadzu UV-2600i, S/N: A12595900733

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(Last edited: TCrockford June 18, 2026)

Dissolved absorption spectra (a_d)

I. Sampling & Running a_d:

1. Collect surface water into 2 clean, acid washed 4oz amber glass bottles.
2. Set up an all-glass flask and filtration apparatus (e.g., Millipore Corp.) with a 0.22 µm polycarbonate filter. Glassware should be previously cleaned in sequence with 2% Micro-90 (International Products Corporation), 10% HCl, and Milli-Q water (Millipore). See photo below.



3. Filter approximately 100 mL of fresh Milli-Q water at 5 inches Hg vacuum pressure through the new filter and discard filtrate. (Perform this step each time a new filter is used.)
4. Filter approximately 40 mL of the first sample bottle. Swirl filtrate in flask to rinse flask and then use filtrate to rinse a new acid-washed amber sample bottle. Discard filtrate after rinsing bottle. Repeat this step for a total of 2 rinses.
5. Filter the rest of the sample (with 5 inches Hg vacuum pressure) and store in the rinsed glass bottle in the fridge for up to 6 months.

*The 0.22 µm polycarbonate filter can be reused if filtering multiple samples the same day (starting at step 4 for each sample), however filtering will eventually become so slow that you will want to change the filter. Typically, one filter will last for 2-3 samples.

II. Running a_{cdom}

1. Samples are run on a dual beam spectrophotometer with standard optics and cylindrical quartz cuvettes, type 34, path length 10mm, match code 6 (Starna Cells, Inc.).
 - Cuvettes are previously washed in 2% Micro-90, 10% HCl, soaked in Milli-Q water, and stored dry.
2. If samples have been refrigerated, allow them to warm to room temperature out of the light (keep dark with aluminum foil).
3. Turn on the spec to warm up for 20 minutes before starting.
4. Set spectrophotometer scan settings to spec-specific lab protocol:
 - Perkin Elmer Lambda 18
 - Range: 200-900 nm
 - Resolution: 1 nm
 - Scan Speed: 240 nm/min
 - Smooth: 6 nm
 - Beam slit: 4.00 nm
 - Lamp UV & Lamp Vis: ON
 - Lamp change: 319.2 nm
 - Ordinate max/min: 1.000/-0.010
 - Perkin Elmer Lambda 650 S (Serial # 650L1806192)
 - Range: 200-900 nm
 - Resolution: 1 nm
 - Scan Speed: "slow speed"
 - Smooth: none
 - Beam slit: 4.00 nm
 - Lamp UV & Lamp Vis: ON
 - Lamp change: 326 nm
 - Shimadzu UV-2600i, S/N: A12595900733
 - Range: 200-900 nm
 - Resolution: 1 nm
 - Scan Speed: "slow speed"
 - Smooth: none
 - Beam slit: 5.00 nm
 - Lamp UV & Lamp Vis: ON
 - Lamp change: 319 nm
5. Run autozero/baseline and scan air vs. air (nothing in sample or reference light path) to check the baseline of the spectrophotometer.
6. Check that the faces of the quartz cuvettes are clean and free of smudges before every scan. If dirty, clean with methanol and lens paper.
7. Fill both cuvettes with Milli-Q water, taking care to avoid bubbles. Place one in sample beam and one in reference beam.
8. Run autozero/baseline and scan Milli-Q water vs. Milli-Q water to check the baseline for subsequent samples.
9. Rinse sample cuvette three times with small amount of filtered sample. Carefully fill cuvette with sample so that there are no bubbles.
10. Run scan of sample vs. Milli-Q water.